Mesa Examples Repository Audit Report

Repository: mesa-examples

Audited by: Tushar Mahato

Audit date: March 2026

Mesa version used: Mesa 3.x / development version

Environment: Python virtual environment with Solara-based visualization.

1. Objective

The goal of this audit was to evaluate the current health of the mesa-examples repository.
The audit focused on the following aspects:

Installation clarity

Dependency specification

Execution instructions

Compatibility with the latest Mesa version

Runtime errors

Documentation completeness

Each example was executed using the instructions provided in its README.

2. Methodology

For every example the following criteria were evaluated:

Installation instructions clarity

Required dependencies

Execution command

Execution result

Errors encountered

Category of issue

Examples were categorized into:

Stable

Legacy visualization

Documentation issue

Packaging/import issue

API incompatibility

Dependency issue

Visualization compatibility issue

3. Example Evaluation Table

Example Evaluation Table			
Example	Result	Category	Issues
aco_tsp	Runs	Mostly Stable	Occasional random runtime error
boltzmann_wealth_network	**Runs**	**Stable**	**No Issues**
caching_and_replay	**Fails**	**Legacy visualization**	**Uses removed mesa.visualization**
charts	**Fails**	**Legacy visualization**	**Depends on CanvasGrid**
color_patches	**Fails**	**Legacy CLI**	**Uses mesa runserver**
conways_game_of_life_fast	**Fails**	**Visualization compatibility**	**Unknown space type NoneType**
deffuant_weisbuch	**Runs**	**Stable**	**Minor matplotlib warning**
dining_philosophers	**Runs**	**Stable**	**No Issues**
el_farol	**Not executable**	**Documentation issue**	**Only notebook instructions**
emperor_dilemma	**Fails**	**Packaging issue**	**Relative import error**
hex_ant	**Fails**	**Packaging issue**	**Relative import error**
hex_snowflake	**Fails**	**Legacy CLI**	**mesa runserver**
hotelling_law	**Runs**	**Stable**	**No Issues**
humanitarian_aid_distribution	**Runs**	**Documentation Issue**	**Missing dependency instructions**
rumor_mill	**Runs**	**Stable**	**Good Documentation**
shape_model	**Fails**	**Legacy Visualization**	**Uses old Mesa server**
termites	**Fails**	**API incompatibility**	**PropertyLayer API change**
virus_antibodies	**Runs**	**Documentation Issue**	**Installation unclear**
warehouse	**Runs**	**Stable**	**Well Structured**
forest_fire	**Runs**	**Dependency Issue**	**Missing networkx in requirements**

Summary Statistics

Total examples audited: 20

Category	Count
Stable	7
Legacy Visualization	5
Documentation Issues	3
Packaging/Import Issues	2
API incompatibility	1
Dependency Issues	1
Visualization compatibility issues	1

5. Major Observed Problems

5.1 Legacy Visualization System

Several examples rely on the deprecated Mesa visualization API.

Common indicators:

- mesa runserver
- CanvasGrid
- server.py
- run.py

These examples fail because modern Mesa uses SolaraViz.

Affected examples:

- charts
- shape_model
- caching_and_replay
- hex_snowflake
- color_patches

5.2 Missing or Incomplete Documentation

Multiple examples lack clear setup instructions.

Typical issues:

- missing dependency lists
- unclear installation commands
- incomplete execution instructions

Affected examples:

- humanitarian_aid_distribution
- virus_antibodies
- el_farol

5.3 Packaging and Import Errors

Some examples fail due to incorrect Python package structure.

Error example:

ImportError: attempted relative import with no known parent package

- Affected examples:
- emperor_dilemma
- hex_ant

5.4 API Compatibility Issues

Some examples rely on older Mesa APIs that have changed.

Example:

add_property_layer() signature changed

Affected example:

Termites

5.5 Dependency Management Issues

Some examples fail because dependencies are not declared.

Example:

forest_fire requires network but does not specify it.

6. Overall Repository Health

The audit shows that the repository currently suffers from example bit rot.

Only 7 out of 20 examples run successfully without issues.

Key causes include:

- evolution of Mesa APIs
- migration from Mesa server to Solara
- missing dependency specifications
- inconsistent documentation standards

7. Key Recommendations

To restore the repository's usefulness as a learning resource:

1. Standardize example structure

   Each example should include:

   - README.md
   - requirements.txt
   - run command
   - dependency instructions

2. Replace legacy visualization

   Convert old examples using:

   mesa runserver

   CanvasGrid to SolaraViz.

3. Add automated testing

   CI should periodically:

   install dependencies

   run examples

   detect breakage

4. Add dependency specification

Every example should provide:

requirements.txt or environment.yml

5. Improve documentation consistency

Each README should include:

- installation
- dependencies
- run command
- expected output

8. Conclusion

The mesa-examples repository remains a valuable resource, but many examples are currently broken or outdated due to changes in the Mesa framework.

Addressing these issues through documentation improvements, API updates, dependency management, and automated testing would significantly improve the usability and maintainability of the repository.